Santhosh BV
Electrical Engineering
Indian Institute of Technology Bombay

24B1290
B.Tech.
Gender: Male
DOB: 26/12/2006

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	9.88 **Core:** 136 **| Total:** 136
Intermediate	CBSE	Sri Chaitanya Techno School	2024	98.20%
Matriculation	CBSE	Sharanalaya International School	2022	99.00%

Pursuing a Minor Degree in **Machine Learning & Data Science** from C-MInDS IITB

Scholastic Achievements

- Recognized with the **Institute Academic Prize** for academic excellence in the first year among **1300+** students *('25)*
- Received **Academic Proficiency** in Programming for **Data Science**, sole awardee in a class of **200+** students *('26)*
- Secured **Academic Proficiency** in Microprocessors & Introduction to **Economics** in a class of **120+** students *('25)*
- Bagged a perfect **Semester Performance Index** (SPI) of **10** in the second semester at IIT Bombay *('25)*
- Received **Academic Proficiency** grade in **Linear Algebra & Differential Equations** among **330+** students *('25)*

Key Technical Projects

Financial Derivatives & Time Series Analysis — *May '26 - Jul '26*
Reading Project: Summer of Quant |Quant Club, IIT Bombay

- Developed a comprehensive understanding of **financial derivatives**, their **pricing** using no-arbitrage, Black–Scholes, binomial trees, cost-of-carry models, and their application in hedging and risk management through futures and swaps
- Analyzed time series using **ARIMA**, GARCH models and employed diagnostic tests for volatility modeling, forecasting

Quantitative Trading Strategy & Risk Analysis — *Dec '25*
Quant101 | Finance Club, IIT Bombay

- Designed, and backtested a trend-following momentum trading strategy using **MACD** and **RSI** on **NIFTY-50** spot data, delivering a **17.57% CAGR** over five years, significantly outperforming the **13.25%** buy-and-hold benchmark
- Quantified portfolio risk by estimating 1-day 99% VaR as 2.26% (Historical Simulation) and 1.99% (Variance–Covariance)

Pairs Trading Using Kalman Filtering — *Dec '25 - Present*
WiDS Accio Alpha | Analytics Club, IIT Bombay

- Explored univariate and multivariate **time series analysis** techniques, implementing stationarity tests including ADF, Hurst Exponent, alongside cointegration tests like Johansen and CADF for pair-trading regression analysis
- Analyzed Kalman Filter state-space modeling, prediction/update cycles, noise covariance tracking for optimal estimation

Object Localisation using Wildlife Images — *Oct '25 - Nov '25*
Course Project: Programming for Data Science | Prof. Vinay Kulkarni, C-MInDS

- Achieved a **0.85 Test F1-score** using an **XGBoost** classifier for wildlife localization, leveraging **47 handcrafted** features including Gabor, Sobel filters, saliency maps, HSV analysis, **block coordinates** to capture spatial context
- Included spatial descriptors using the **mean** and **standard deviation** of values within the cells and their neighbors
- Divided image into **8*8** grid, performed semi-automated labeling via clustering, manually refined for ground truth

Sentiment Analysis — *Oct '25 - Nov '25*
NLP Bootcamp | Analytics Club, IIT Bombay

- Performed sentiment analysis on a large-scale customer reviews dataset, achieving a **0.83 F1-score** by implementing an **LSTM** architecture with **128-dimensional** embedding and hidden layers using pyTorch for sequence modeling
- Trained the model on dictionary built after **lemmatizing** and removing stop words until BCE loss was minimized

Customer Segmentation & CLV Prediction — *Sep '25*
Self Project

- Achieved **0.93** $\mathbf{R^2}$ in predicting customer lifetime value by modeling **400K+** invoices using RFM, AOV, and Tenure
- Preprocessed data and customer-level features to apply **Gaussian Mixture Model**, deriving 7 actionable clusters
- Evaluated ML models, showed Gradient-Boost delivered highest accuracy in CLV prediction with key revenue insight
- Applied CLV prediction to customer clusters to optimize marketing, customer satisfaction and operational efficiency

Probabilistic Risk & Premium Analysis
Sep '25

Course Project: Probability and Random Processes | Prof. Nikhil Karamchandani, Department of Electrical Engineering

- Developed a **probabilistic modeling** framework to estimate break even **premium** charges, utilizing **mortality rate** datasets, average income and assuming that the policy purchases per age group followed a **Poisson** distribution
- Designed a premium determination framework with constraints on **minimum surplus margins**, applying the thinning property of Poisson, **cross-subsidization** concepts, and **central limit theorem** for realistic adjustments

Strategic Planning & Analysis
Sep '24 - Nov '24

Course Project: Introduction to Management | Prof. Mayank Pareek, Shailesh J. Mehta School of Management

- Collaborated with a **team of ten** to develop strategic plans for Nestle, analyzing mission, vision, values, and conducting a SWOT analysis to create effective strategies for long-, short-term, **value chain** optimization and **budget** allocation
- Analyzed industry competition using **Porter's Five Forces**, **VRIO** framework to assess leverage of Nestle's products

Other Projects

Computer Architecture
May '26 - Present

Summer of Science | Maths and Physics Club, IIT Bombay

- Analyzing modern processor architectures, combining out-of-order superscalar execution using **Tomasulo's algorithm**, reservation stations and reorder buffers with **cache hierarchy**, memory-system optimization and branch prediction
- Exploring GPU architectures, SIMD, vector processors to understand **data-level parallelism** for parallel computing

Embedded Sound Experimentation System
Jan '26 - Apr '26

Course Project: Design Thinking for Innovation | Prof. Nishant Sharma, IDC

- Developed in a team of 6 a **sound experiment module**, applying the Double Diamond **design thinking** process
- Integrated a Raspberry Pi Pico, INMP441 microphone, servos, and a Streamlit web interface to build an **embedded system** for sound experiments,demonstrating frequency, resonance, and timbre through real-time sensing, actuation

8051-Based IMD Frequency Analyzer
Apr '26 - May '26

Course Project: Microprocessors | Prof. Shalabh Gupta, Department of Electrical Engineering

- Optimized memory usage for selective spectral analysis on **8051** by replacing the DFT with the **Goertzel algorithm**
- Developed synchronized ADC/DAC operation using **single timer** to reduce jitter, **phase-accumulator** LUT-based DDS, quantization-aware frequency selection, and an integer square-root algorithm for IMD amplitude computation

CPU Design in Verilog
Oct '25 - Nov '25

Course Project: Digital Systems | Prof. Sachin B Patkar, Department of Electrical Engineering

- Designed and implemented an **8-bit single-cycle** Harvard CPU in Verilog supporting **23** instructions across various categories, including data movement, memory access, arithmetic, logical, relational, and complex control flow operations
- Extended the architecture to a **16-bit multi-cycle** Von Neumann CPU, redesigning the datapath to integrate FSM-based control logic and validated the design by executing **Bubble Sort**, confirming correct instruction sequencing

I^2C-Based Multi-Sensor Interface
Apr '26

Microprocessors Laboratory | Prof. Sachin B Patkar, Department of Electrical Engineering

- Developed a reusable **I^2C driver** on the 8051 based AT89C5131A to interface BMP280- temperature sensor, and DS1307 RTC modules, enabling synchronized communication with multiple peripherals over a shared **two-wire bus**
- Implemented real-time processing of 20-bit calibrated sensor data with concurrent LCD and **UART** communication

Technical Skills

Programming	C++, Python, Verilog, VHDL, Embedded-C, 8051 Assembly
Softwares and Packages	Quartus, LTspice, EasyEDA, Keil μVision, Fusion360, Fracktory, LaTeX
Python Libraries	NumPy, Pandas, Matplotlib, SciPy, Seaborn, Sklearn, Statsmodels, pyTorch, OpenCV

Extracurriculars

- Attended a 5-day bootcamp on **communication**, **teamwork**, and leadership conducted by Career Cell, IITB ('25)
- Built an **RC plane** using Depron, powered by BLDC motors with ESC and servo controls for pitch and roll. Competed and recorded fly time of **10+s** in the RC Plane Competition organized by the Aeromodelling Club, IIT Bombay ('24)
- Participated in building an autonomous **Mini Militia bot** for Code Wars organized by WNCC, IIT Bombay ('26)
- Selected from over **1000+** students, in the first year, for year-long **basketball** training under **NSO**, IIT Bombay ('25)
- Participated in the department basketball competition organized by Electrical Engineering Students' Association ('24)
- Designed a multipurpose pen stand, optimized **hostel room organization** in the Introduction to Design course ('24)